

Estimating latent factors in dynamic models

The Kalman filter under misspecification

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Introduction

What is the Kalman filter (Hamilton, 1994; Kalman, 1960)?

Core idea

Estimate a dynamic latent factor by combining the model's prediction with new information

Consider the following dynamic factor model:

$$y_t = \Lambda f_t + e_t$$

What is the Kalman filter (Hamilton, 1994; Kalman, 1960)?

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Consider the following dynamic factor model:

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The Kalman filter estimates the latent factor with a **two-step recursion**:

1. **Prediction:** forecast the factor at time t using information until time $t - 1$
2. **Update:** correct the forecast using new observations at time t

Central question

How does the Kalman filter perform when its assumptions are violated?

We study **two types of misspecification**:

1. **Distributional**: shocks are non-Gaussian (heavy-tailed, skewed).
2. **Structural**: the number of latent factors is misspecified.

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Starting point: co-movements across macroeconomic indicators are driven by a *single* latent factor, the business cycle (Stock & Watson, 1988).

The Stock & Watson dynamic factor model

Four indicators and one single factor, with AR(2) dynamics for both the factor and the errors.

Measurement equation: $\mathbf{y}_t = \mathbf{\Lambda} f_t + \mathbf{e}_t$

Factor transition equation: $f_t = \phi_1 f_{t-1} + \phi_2 f_{t-2} + w_t, \quad w_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$

Error transition equation: $e_{i,t} = \psi_{i,1} e_{i,t-1} + \psi_{i,2} e_{i,t-2} + \varepsilon_{i,t}, \quad \varepsilon_{i,t} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_i^2)$

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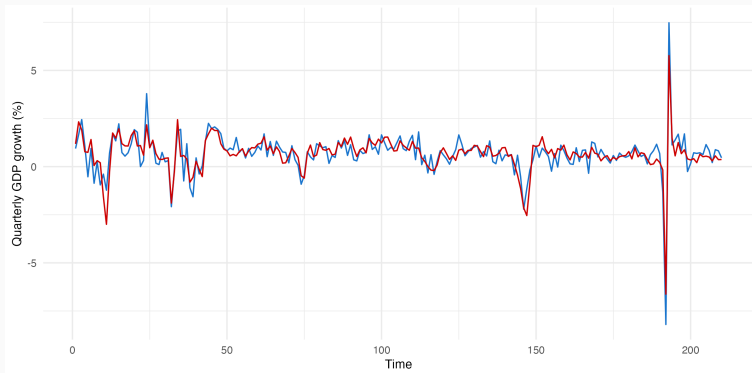
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This model encodes two **conditional independence statements**:

1. **Contemporaneous:** $y_{i,t} \perp\!\!\!\perp y_{j,t} \mid f_t \quad \forall i \neq j$
2. **Over time:** $y_t \perp\!\!\!\perp y_s \mid h_t \quad \forall t \neq s$, where $h_t = (f_t, f_{t-1}, e_{1,t}, e_{1,t-1}, \dots, e_{N,t}, e_{N,t-1})$

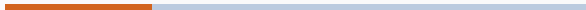
Replication of the Stock & Watson model

Actual (blue line) versus fitted (red line) GDP growth:



- Factor tracks GDP growth well ($R^2 = 0.763$ and $MSE = 0.276$).
- All loadings on the factor are positive and between 0.525 and 0.926.

Breaking Gaussianity



Chapter 2: Breaking the Gaussianity Assumption

Setup: simulate the S&W DGP under three distributions of increasing complexity; evaluate factor recovery across sample sizes.

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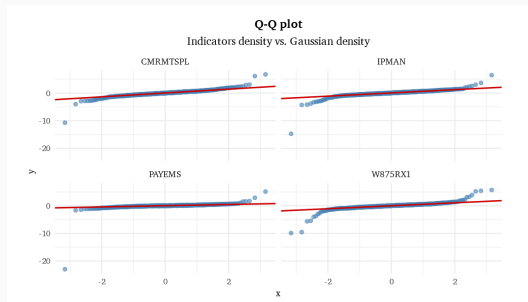
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Theoretical prediction (QMLE)

The Kalman filter is **consistent** for all three distributions (White, 1982), but loses **efficiency** in finite samples as the distribution departs from Gaussianity.

Chapter 2: Macro data is not Gaussian



Q-Q plots: fat tails correspond to points deviating from the 45° line at the extremes.

- (Geweke, 1993) finds posterior odds favor Student- t over Gaussian for 13/14 macro indicators
- Even at $\nu = 5$, evidence against normality is overwhelming

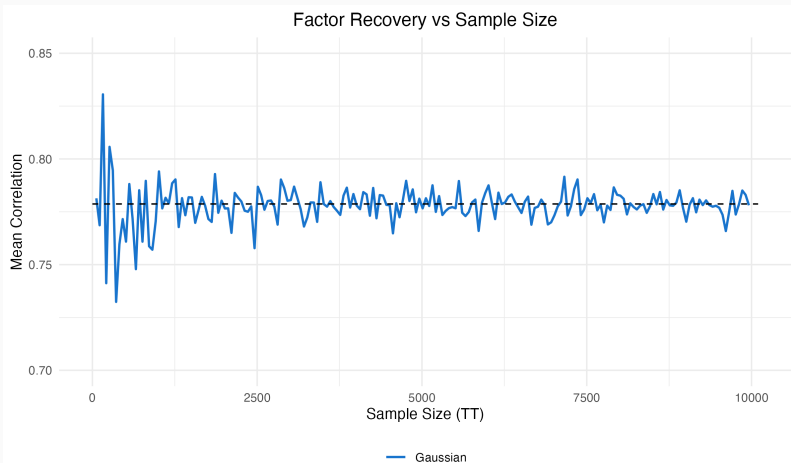
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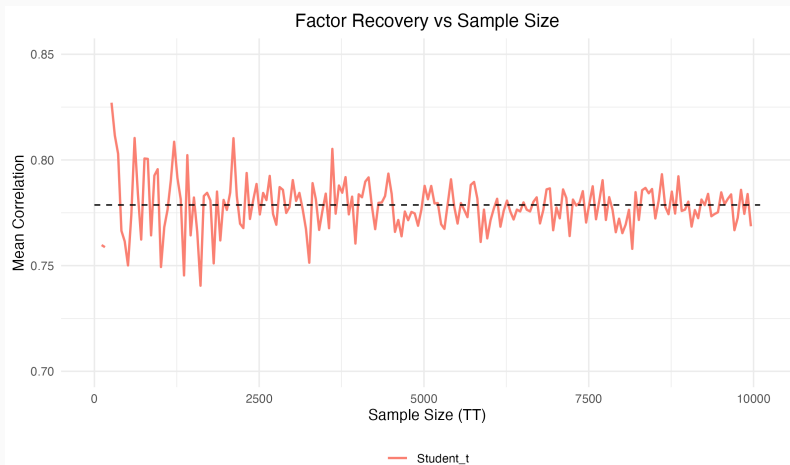
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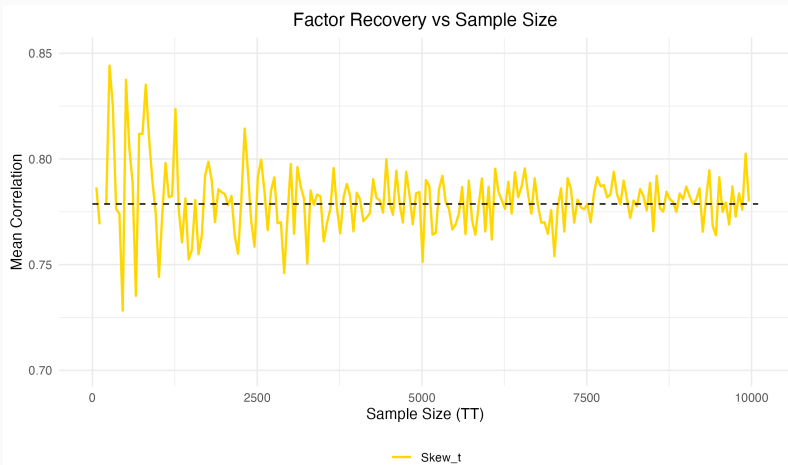
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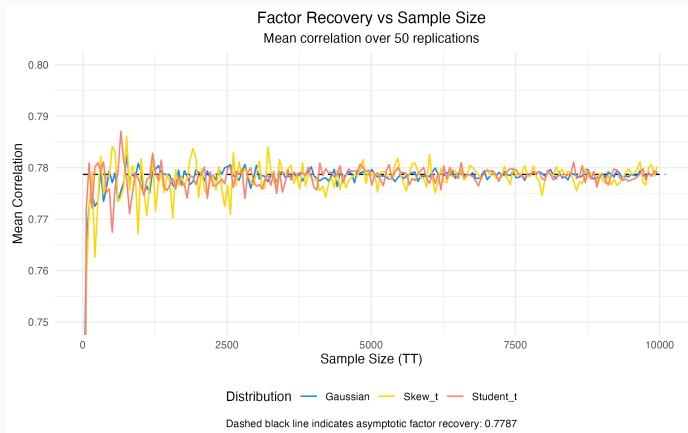
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Chapter 2: Monte Carlo Results (50 replications)

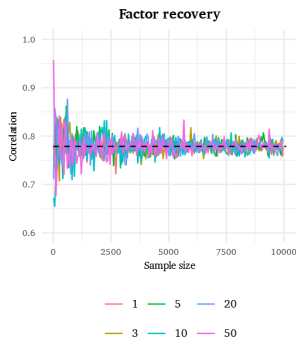
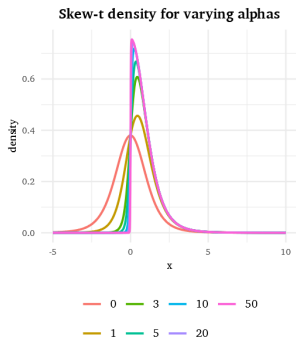


Factor recovery at $T = 633$

Distribution	Mean	SD
Gaussian	0.777	0.016
Student- t	0.778	0.025
Skew- t	0.777	0.030

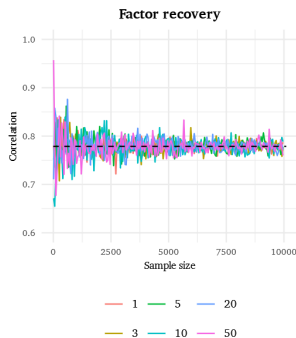
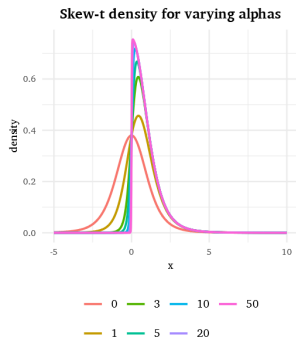
Same mean, increasing variance $\Rightarrow$
consistency without efficiency.

Chapter 2: (extension) How extreme can the skew get?



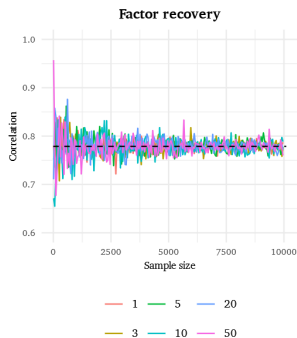
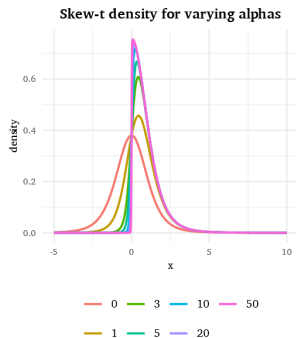
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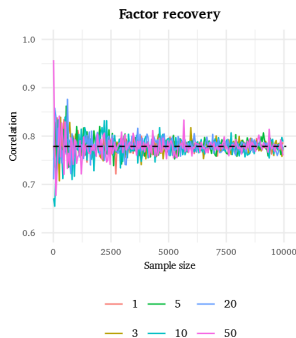
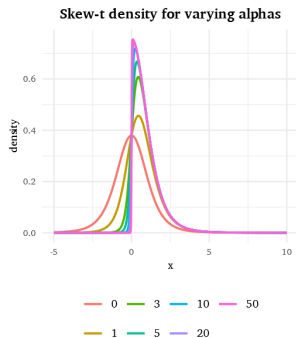
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QMLE consistency is robust even to extreme skewness.

Factor misspecification



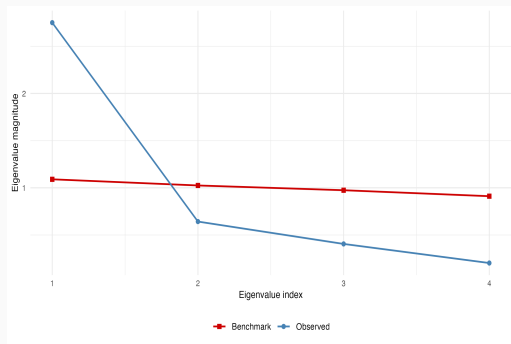
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- Run parallel analysis to estimate the number of factors (Horn, 1965).

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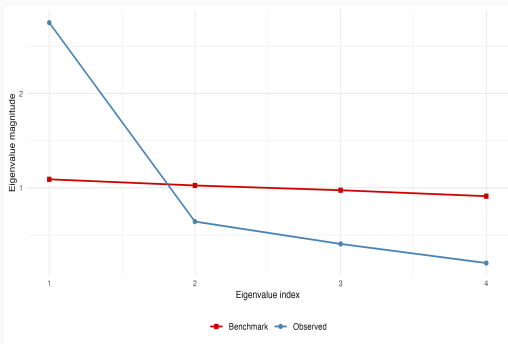
Original model



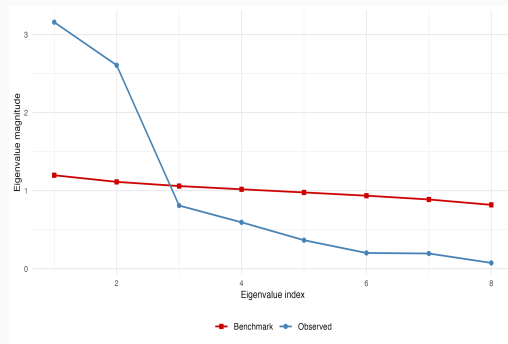
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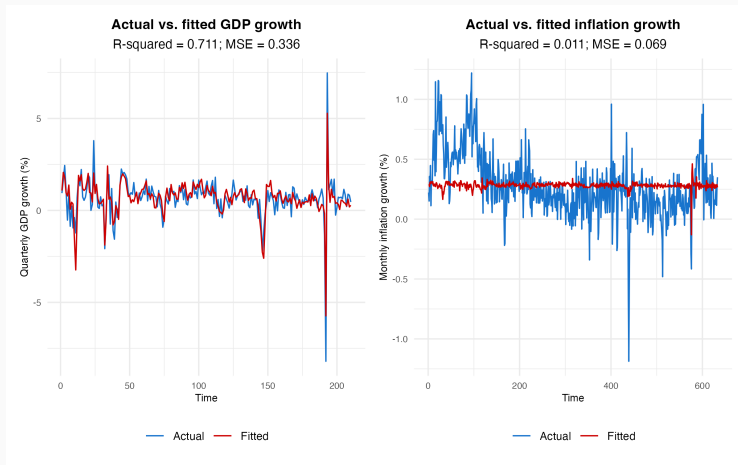
Augmented model



Note: The benchmark eigenvalues have been simulated from an AR(2) noise process with 500 Monte Carlo replications.

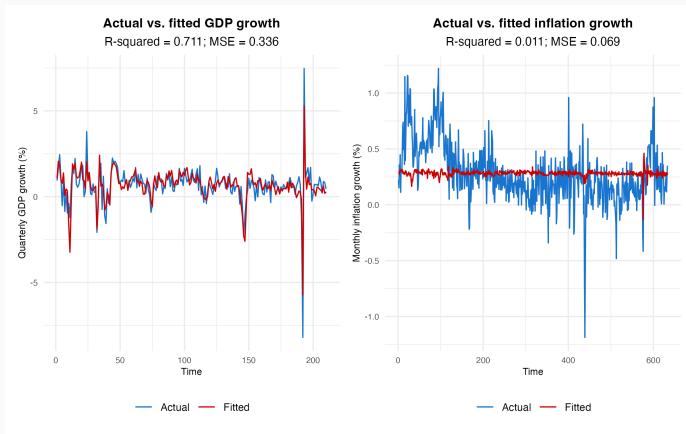
Estimation under misspecification of the number of factors

If the number of factors is misspecified, the Kalman filter recovers only one of them.



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Avg. Loadings ($\hat{\lambda}$)

Group	Average
Real Activity	0.723
Prices	0.110

Result: The Kalman filter captures the business cycle and discards inflation information.

Why does this happen?

To explain this, we can draw a parallel to Probabilistic Principal Component Analysis

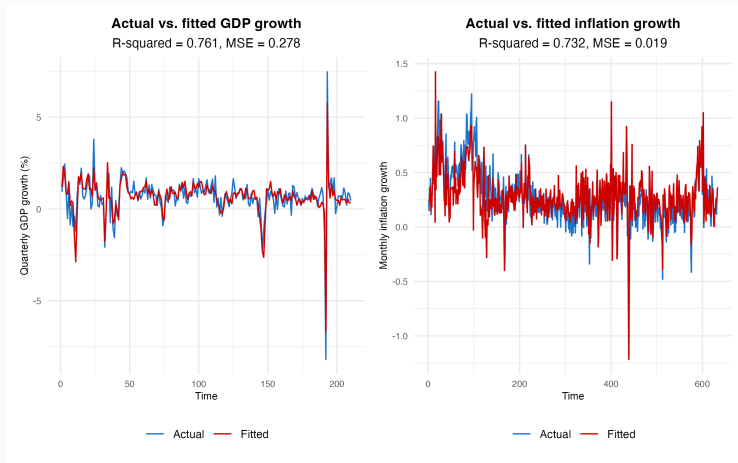
Under PPCA, the MLE of the loading vector with only one factor is (Tipping & Bishop, 1999):

$$\lambda_{\text{ML}} = \mathbf{u}_1 (d_1 - \sigma^2)^{1/2}$$

- The **direction** is determined by $\mathbf{u}_1$, *leading eigenvector* of the sample covariance matrix.
- The eigenvector $\mathbf{u}_1$ is associated to the *largest eigenvalue*.
- The largest eigenvalue is the one associated to the **real activity** factor.

Correctly specified model

With the correct two factor specification, the Kalman filter recovers both factors.

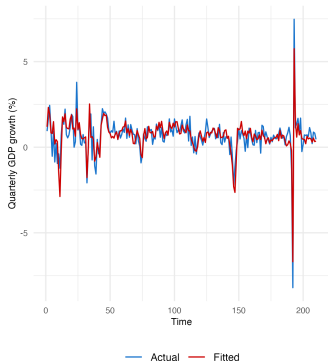


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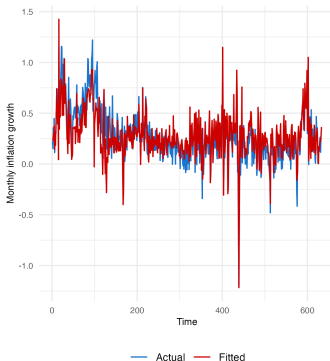
Actual vs. fitted GDP growth

R-squared = 0.761, MSE = 0.278



Actual vs. fitted inflation growth

R-squared = 0.732, MSE = 0.019



Avg. Loadings ($\hat{\lambda}$)

Group	Average
-------	---------

GDP Factor

Real Act.	0.735
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Prices	0.108
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Inflation Factor

Real Act.	-0.073
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Prices	0.513
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Conclusion and references

Main findings:

- **Gaussianity:** Kalman filter is **consistent** under non-Gaussian shocks, but loses efficiency as distributions become more elliptical or skewed.
- **Factor misspecification:** a misspecified single-factor model captures only the **dominant signal**, discarding the rest.

Going forward:

- Does adding more indicators improve performance under correct model specification?
- Allowing for time-varying loadings would more closely mirror the empirical application

References

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